■ SingularValues

`SingularValues[`m`]` gives the singular value decomposition for a numerical matrix m. The result is a list `{`u`,` w`,` v`}`, where w is the list of nonzero singular values, and m can be written as `Transpose[`u`].DiagonalMatrix[`w`].`v.

`SingularValues[`m`, Tolerance -> `t`]` specifies that singular values smaller than t times the maximum singular value are to be removed. ■ The default setting `Tolerance -> Automatic` takes t to be 10 times the numerical precision of the input. ■ u and v can be considered as lists of orthonormal vectors. ■ See page 454. ■ See also: `PseudoInverse`.

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